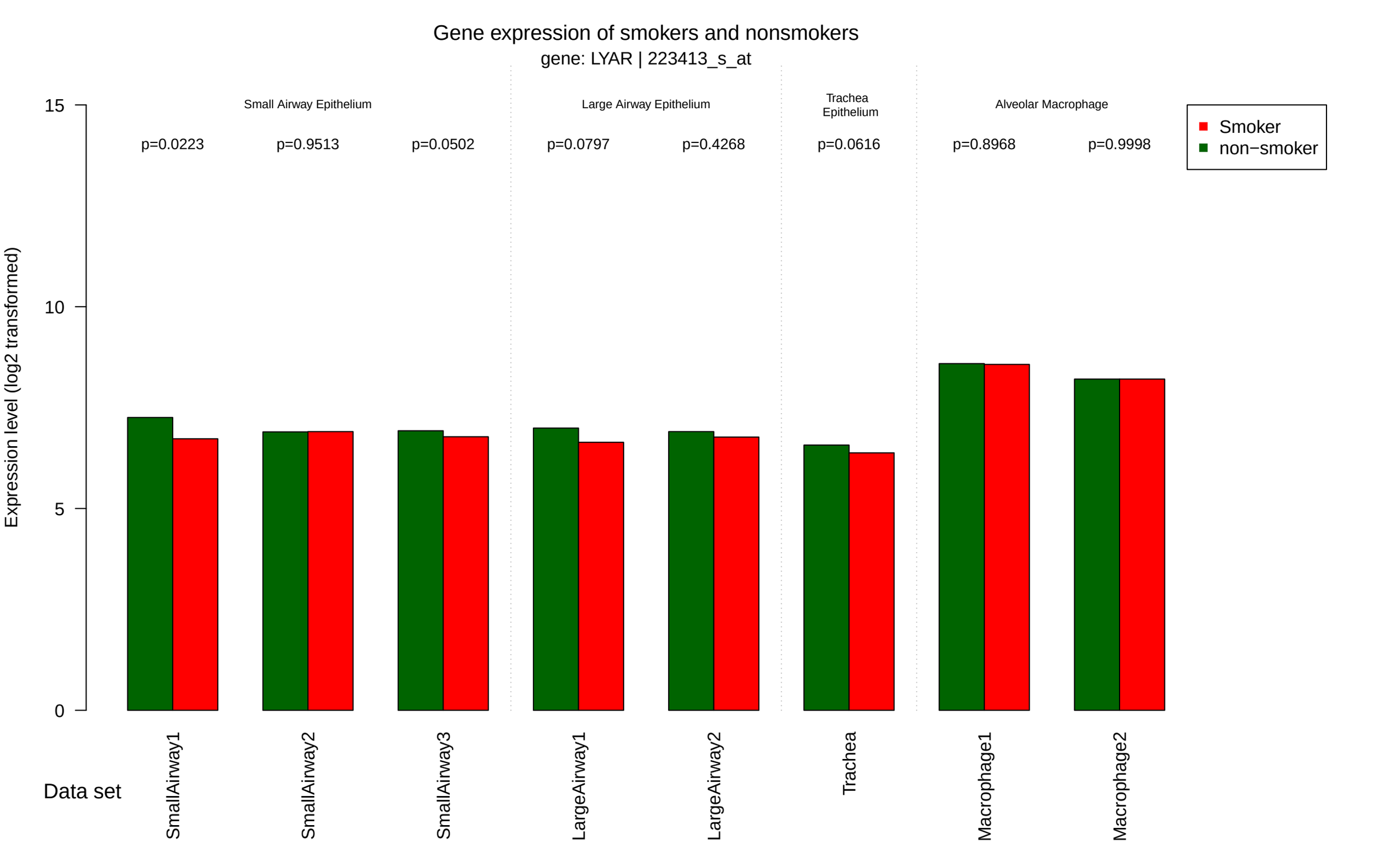
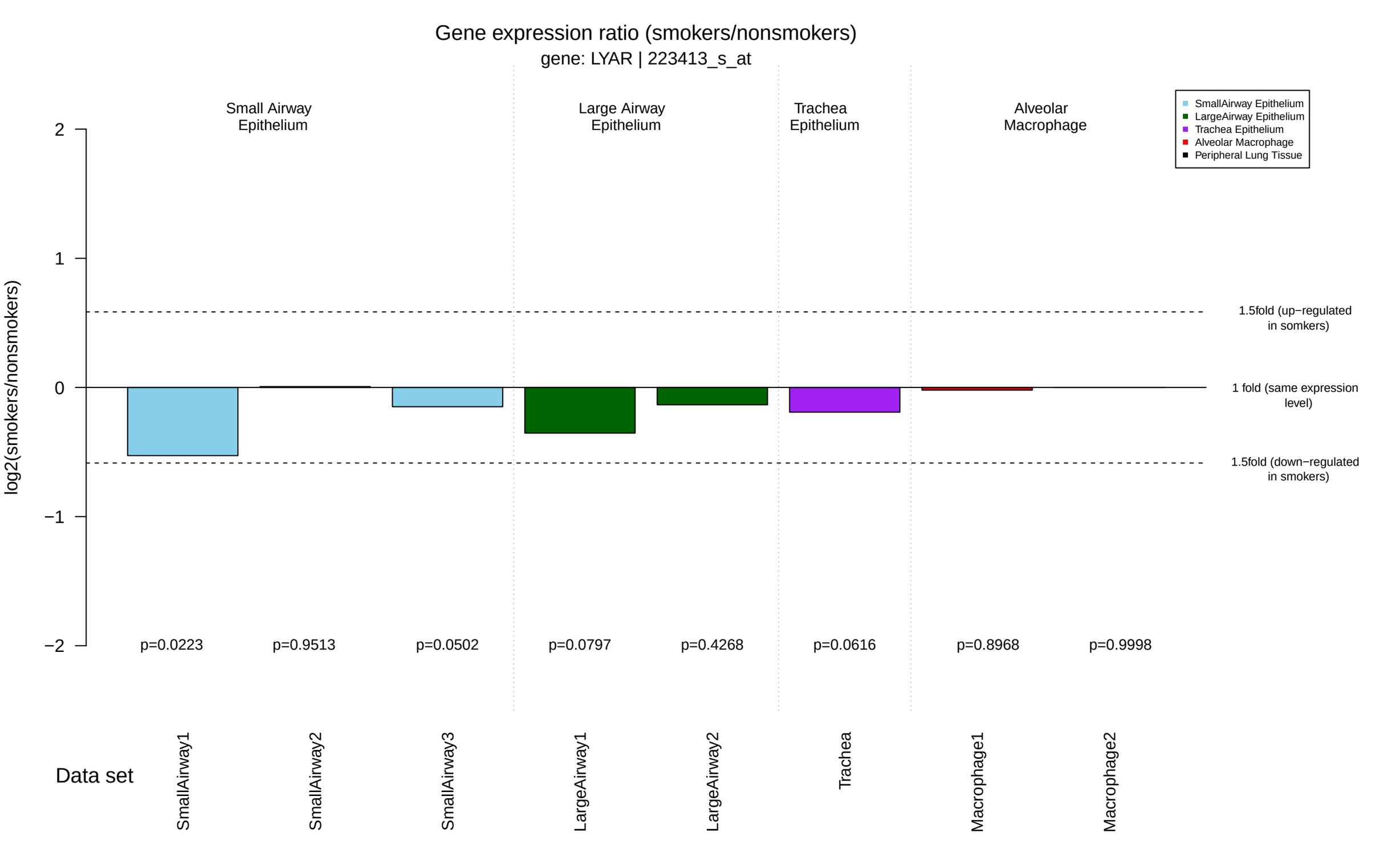
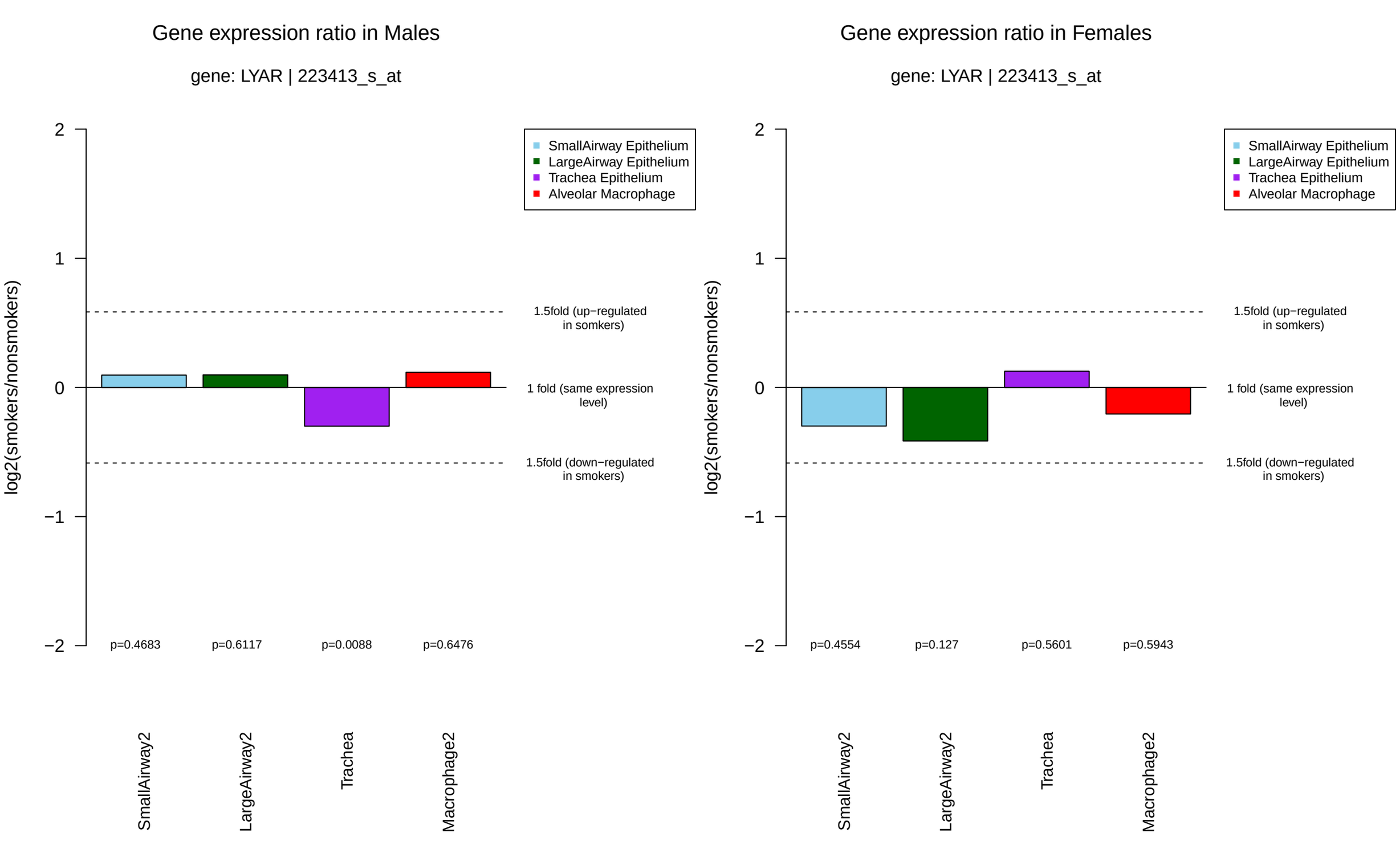
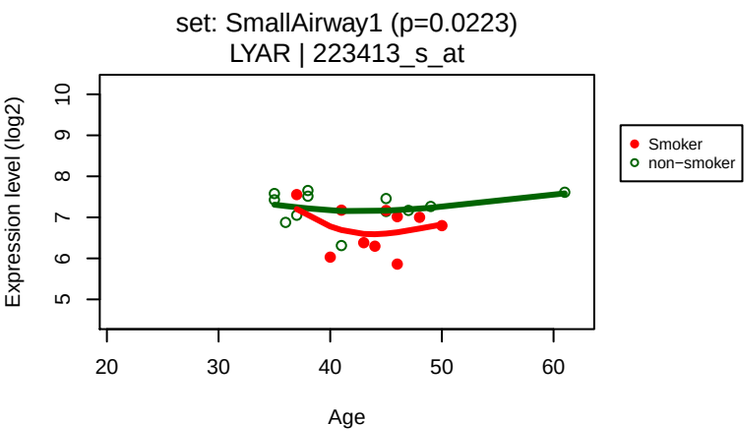
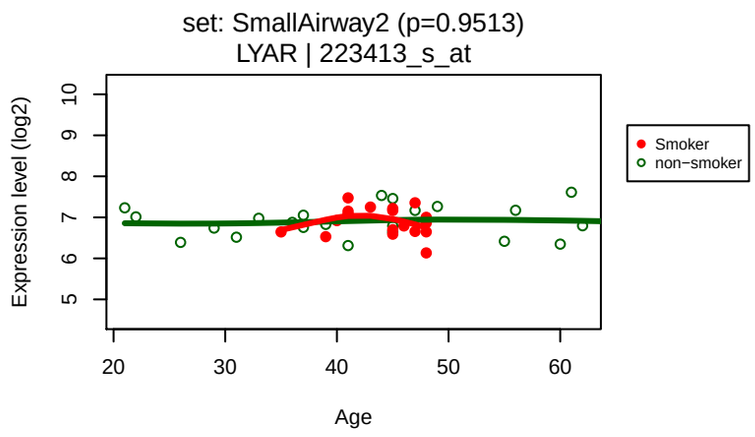
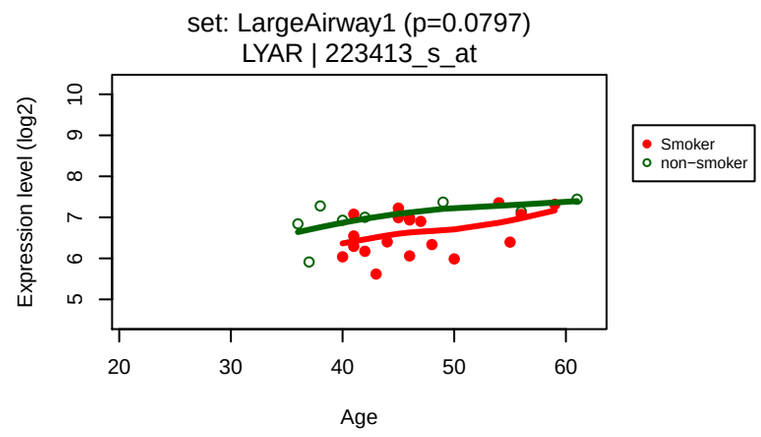

Gene expression of smokers and nonsmokers
gene: LYAR | 223413_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0223
p=0.9513
p=0.0502
p=0.0797
p=0.4268
p=0.0616
p=0.8968
p=0.9998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

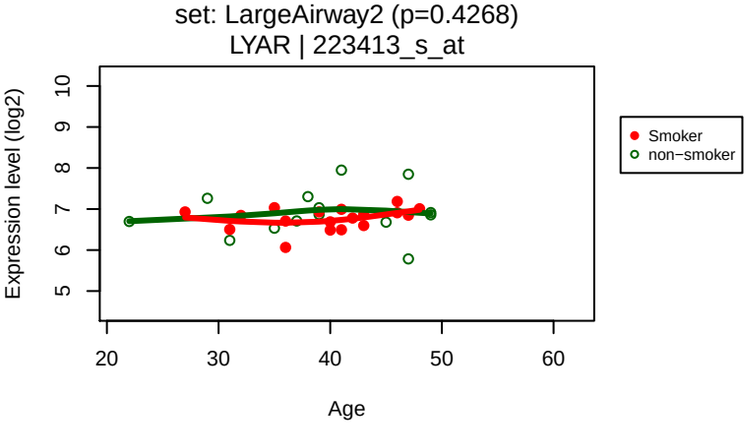

Gene expression ratio (smokers/nonsmokers)
gene: LYAR | 223413_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0223
p=0.9513
p=0.0502
p=0.0797
p=0.4268
p=0.0616
p=0.8968
p=0.9998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

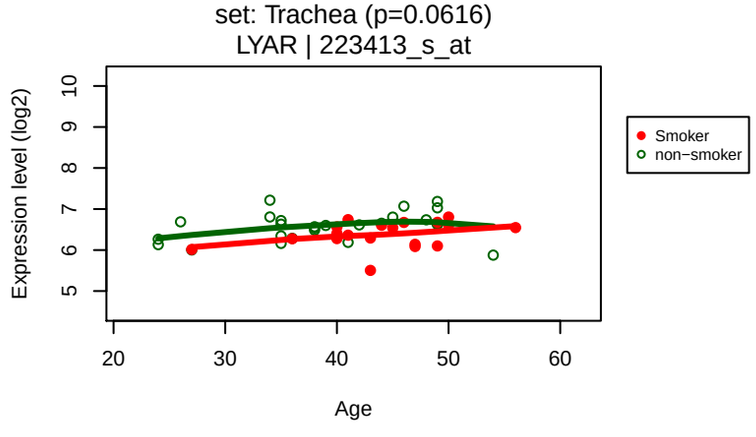

Gene expression ratio in Males
gene: LYAR | 223413_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4683
p=0.6117
p=0.0088
p=0.6476
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LYAR | 223413_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4554
p=0.127
p=0.5601
p=0.5943
SmallAirway2
LargeAirway2
Trachea
Macrophage2

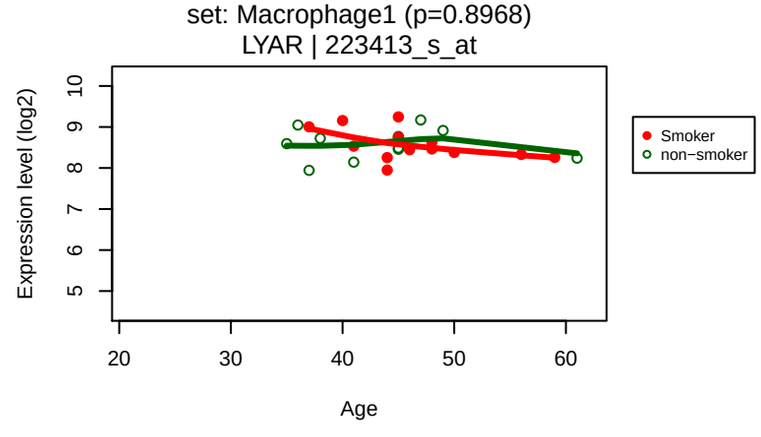

set: SmallAirway1 (p=0.0223)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker

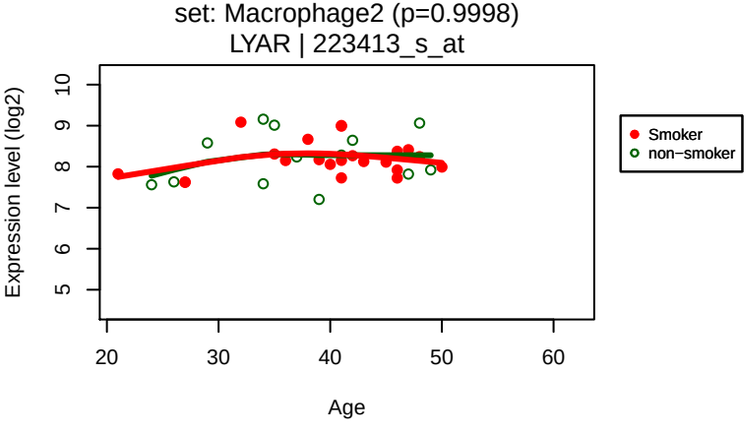

set: SmallAirway2 (p=0.9513)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker

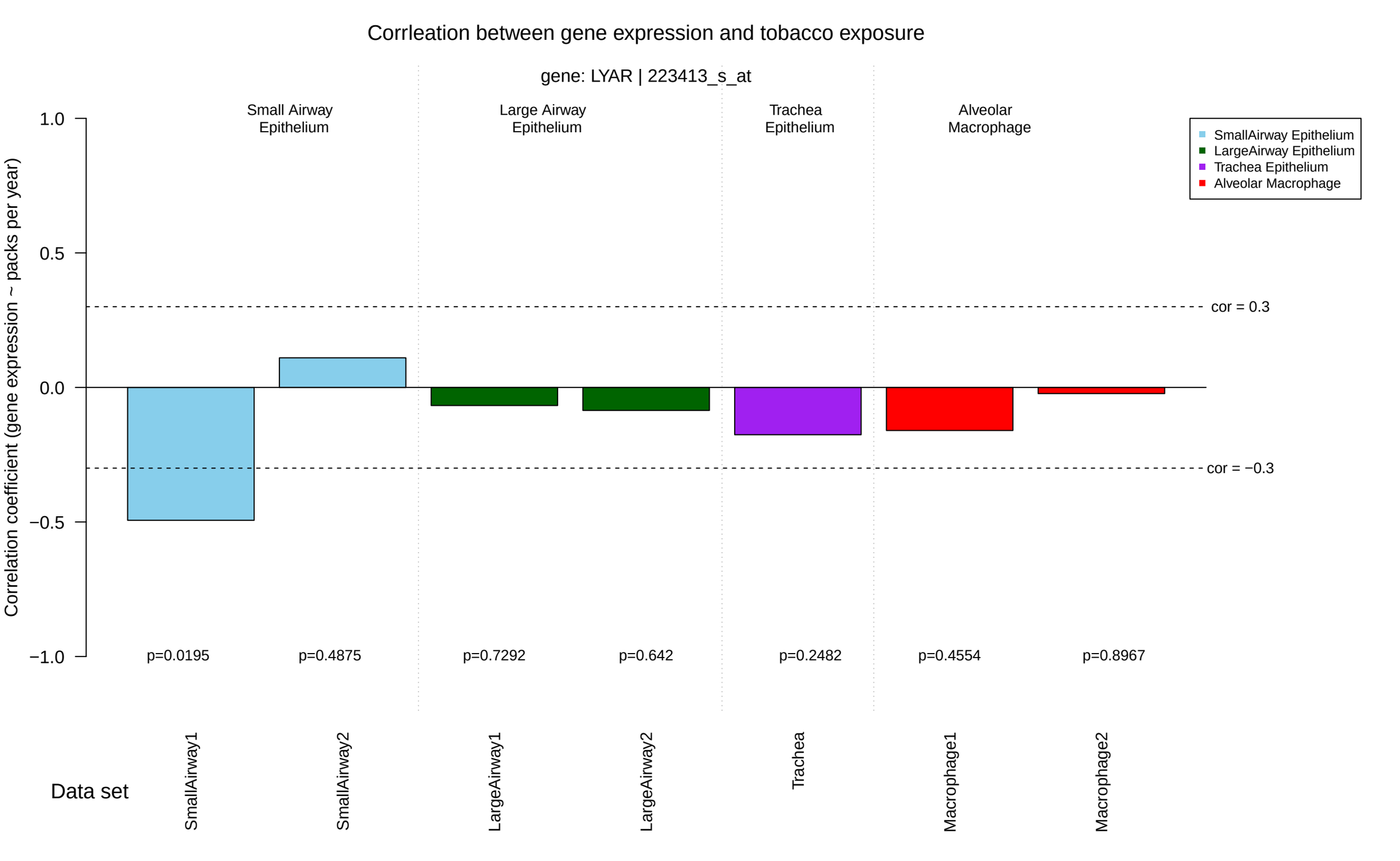

set: LargeAirway1 (p=0.0797)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4268)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0616)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8968)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9998)
LYAR | 223413_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LYAR | 223413_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0195
p=0.4875
p=0.7292
p=0.642
p=0.2482
p=0.4554
p=0.8967
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2